VideoQuiz

AI-Powered Interactive Video Question Generation System

1. Abstract

VideoQuiz is a web application that allows users to upload a video file or provide a video link from a supported source. The system first scans the video and identifies salient information, then automatically generates questions and maps them to specific timestamps in the video. While the video is being played, questions appear at their assigned time slots for the user to answer, followed by feedback and a final performance report.

2. Problem Statement

Educational videos can be useful for learning, but users often watch them passively without checking whether they understood the key information. Creating questions manually for every video is also time-consuming. VideoQuiz solves this problem by turning video content into an interactive learning experience, where relevant questions are automatically generated and displayed during playback.

3. System Overview

The user provides the system with a video file or a supported video URL. The system validates and analyzes the provided content, identifies the important information, generates relevant questions, and assigns timestamps to them. During playback, the system displays each question at its assigned time, records whether the user's answer is correct or incorrect, provides immediate feedback, and presents a summary of the user's overall performance.

Main flow: Video Input → Content Analysis → Question Generation → Timestamp Assignment → Interactive Playback → Answers and Feedback → Final Results

4. User Requirements

4. The system shall provide users with the key functionalities needed to use VideoQuiz for interactive video learning.

4.1 The system shall allow the user to upload a video file. *(Supporting local files lets users who already have a saved video use the system without needing an online link.)*

4.2 The system shall allow the user to enter a link to a video in a supported format, such as a YouTube URL. *(Supporting links lets users reuse existing online videos, such as recorded lectures, without downloading them first.)*

4.3 The system shall allow the user to play the video and answer questions that appear at specified times during playback. *(This interactive playback is exactly the kind of learning experience the system is designed to provide.)*

4.4 The system shall provide feedback after the user submits an answer. *(Immediate feedback lets the user know right away whether they understood that part of the video.)*

4.5 The system shall show the user's final score or performance summary after completing the activity. *(A summary score gives the user a simple measure of how well they understood the video after finishing it.)*

4.6 The system should allow users to review their previous questions and answers. *(Reviewing past attempts lets a user who answered incorrectly try again without rewatching the entire video.)*

5. Functional System Requirements

5. The system shall perform the functions necessary to process video content and provide interactive questions.

5.1 The system shall validate the uploaded video or provided URL before processing it. *(Validating early avoids spending computing power and resources on a video that cannot actually be processed.)*

5.2 The system shall analyze the content of the video. *(Analysis is the process that turns raw video into information the system can use to generate questions.)*

5.3 The system shall extract key information from the analyzed video to use for question generation. *(If all content were treated as equally important, the generated questions would be trivial, so isolating the most relevant content matters.)*

5.4 The system shall generate questions based on the important information extracted from the video. *(This is the core function that automates work that would otherwise be tedious and manual.)*

5.5 The system shall assign a timestamp to each generated question and display it during playback. *(Timing each question to its source moment keeps it anchored to the content the user just watched.)*

5.6 The system should allow users to retry the analysis if processing fails. *(A retry option saves users the time of re-uploading the video from scratch after a temporary failure.)*

6. Non-Functional Requirements

6. The system shall meet quality requirements related to usability, performance, security, compatibility, and error handling, each stated as a measurable, testable target.

6.1 The system shall allow a first-time user to upload a video and start playback in no more than 3 steps, without requiring a tutorial. *(Stating ease of use as a step count makes it measurable, rather than leaving it as a vague impression.)*

6.2 The system shall display each question within 2 seconds of its scheduled time during normal playback. *(A fixed time bound is testable, unlike an open-ended "noticeable delay.")*

6.3 The system shall display an error message explaining the cause (e.g., unsupported format, unreachable link) within 5 seconds of a failed input or processing request. *(A specific, timely message lets the user fix the actual issue instead of guessing what went wrong.)*

6.4 The system shall require user authentication before a user's uploaded videos, links, answers, and results can be accessed, and shall transmit this data over an encrypted (HTTPS) connection. *(Naming the authentication and encryption mechanism turns a general security goal into something verifiable.)*

6.5 The system shall function correctly on the latest two versions of Chrome, Safari, and Edge, and shall support screen widths from 360px to 1920px. *(Naming specific browsers and a screen-width range replaces "commonly used" with something QA can actually test against.)*

6.6 The system should display a clear status message if an external AI or video service is unreachable for more than 10 seconds, and should retain any video that was already partially uploaded. *(Graceful degradation is verifiable when it is tied to a defined time threshold and fallback behavior, rather than left to interpretation.)*